
Learning Spectral Structure In-Context: Fourier Function Class Induction via Transformers

Abstract

In-context learning (ICL) allows Transformer models to perform new tasks by conditioning on input-output examples without updating parameters. While prior work has demonstrated that Transformers can emulate learning algorithms for simple function classes, it remains unclear whether they can induce structured transformations between complex domains, such as time and frequency. In this work, we explore the capacity of Transformers to in-context learn the Discrete Fourier Transform (DFT) by presenting paired time- and frequency-domain signal segments as prompts. We propose a synthetic task setup where each prompt consists of sampled signals and their spectral representations, and the model must predict the frequency structure of a new time-domain query. Our architecture builds on recent advances in patch-based tokenization and decoder-only Transformers to support spectrum regression and frequency classification tasks. We evaluate performance using metrics including frequency recovery accuracy, generalization to unseen frequencies, and prompt efficiency. Our results demonstrate that Transformers can recover spectral structure in context, offering a new perspective on learning structured signal transformations without gradient updates. Our code can be viewed on <https://anonymous.4open.science/r/in-context-learning>.

1 Introduction

In-Context Learning (ICL) has emerged as a powerful paradigm where large Transformer models can perform new tasks by conditioning on example demonstrations, without updating model parameters. ([1]) Foundational work has demonstrated that Transformers can emulate learning algorithms over simple function classes—such as linear regression, sparse regression, and decision trees—purely from sequence-based prompts. This line of research views ICL as meta-learning and has explored both empirical capacity and theoretical underpinnings of Transformers as function learners ([2]).

In parallel with recent advances in in-context learning, a growing body of work has examined how deep neural networks learn functions in the frequency domain. The Frequency Principle (F-Principle), introduced by [3], shows that neural networks tend to fit the low-frequency components of a target function earlier in training, gradually capturing higher-frequency details over time. This spectral bias has been observed across a wide range of architectures (e.g., CNNs, fully connected networks, VGG16), loss functions, and real-world datasets such as MNIST and CIFAR-10. The underlying cause has been linked to the smoothness of standard activation functions, which implicitly steer optimization toward low-frequency solutions. While the F-Principle is well-studied in supervised learning, its implications for in-context learning remain underexplored. In this work, we investigate whether Transformers can recover spectral structures from time-domain examples presented in context, extending the scope of ICL to include structured transformations grounded in Fourier analysis.

While prior work has independently explored ICL and frequency bias in deep networks, their intersection remains underexplored. Notably, it is still unknown whether Transformers can *in-context learn spectral mappings*—i.e., induce discrete Fourier relationships between time-domain and frequency-domain representations. This paper aims to bridge that gap by evaluating whether

Transformers can induce the structure of discrete Fourier transform (DFT) mappings purely from prompts consisting of paired time-frequency segments.

We propose a novel experimental framework where prompts are composed of sampled signal segments and their corresponding DFT representations. Frequency-domain analysis is a cornerstone in signal/image processing and the study of dynamical systems. For a given discrete time-domain function, $\mathbf{x}[n] \in \mathbb{R}^d$, the frequency components are given by its Fourier Transform:

$$\mathbf{X}[k] = \sum_{n=0}^{N-1} \mathbf{x}[n] e^{-i2\pi kn/N} \quad (1)$$

The model receives several such examples and is tasked with predicting the frequency-domain representation of a new query signal segment.

2 Related Work

Recent research on in-context learning has investigated the capacity of Transformers to perform function learning from input-output demonstrations alone, without parameter updates. A central question in this line of work is: *can Transformers be trained to emulate learning algorithms over structured function classes?*

Function Class Induction Through In-Context Learning

Transformer models have demonstrated a remarkable ability to perform in-context learning (ICL), adapting to new tasks through examples without parameter updates [1, 4]. provided pivotal insights by showing transformers can learn simple function classes in context, while [5] and [6] revealed that attention mechanisms implicitly implement gradient-based optimization. [7] investigated transformer limitations across Boolean function classes for more complex functions, showing performance degradation with increasing task complexity. [8] identified a two-phase ICL model: first identifying task structure and then applying this understanding—a framework relevant to our frequency mapping tasks.

ICL for Complex and Discrete Functions. Beyond continuous-valued tasks, recent works have investigated ICL over discrete and high-complexity function classes. [9] showed that Transformers trained on polynomial functions can achieve ICL within the training distribution (ICL1), but fail to generalize out-of-distribution (ICL2). They also studied inverse tasks such as zero-finding. [7] explored ICL over Boolean function classes, finding that Transformers generalize well on low-complexity classes (e.g., conjunctions) but struggle with high-complexity ones like parity. Importantly, they showed that Transformers can adaptively switch between multiple learned algorithms depending on the prompt, providing evidence of meta-algorithmic behavior.

Spectral Learning and Fourier Transforms in Neural Networks

Recent neural architectures have incorporated frequency-domain information primarily as architectural components rather than learning targets. Specialized architectures like [10], [8], and [11] incorporate frequency information for various applications. [12] developed a Fourier complex Transformer decomposing self-attention via discrete Fourier transform, while [13] introduced FourierLearner-Transformers learning spectral representations of positional encodings. However, these approaches require parameter updates to adapt to new frequency patterns, unlike our proposed method.

The Gap: In-Context Spectral Structure Induction

While transformers have demonstrated capabilities in learning various function classes in context [14, 7], and Fourier transforms have been incorporated into neural architectures, the ability to induce mappings between time and frequency domains without parameter updates remains unexplored. Our work bridges this gap by investigating transformers’ capability to induce spectral structures in context, extending function class induction to complex domain mappings while providing practical benefits for adaptive signal processing.

3 Approach

Our approach builds upon recent advances in transformer architectures and in-context learning to develop a framework for learning mappings between time and frequency domains through contextual examples without parameter updates.

Problem Setup

We define the core task as mapping a time-domain signal segment to its spectral representation. Each task is based on a family of synthetic signals, where each signal is generated as a sparse combination of sinusoids with potentially varying frequencies, amplitudes, and noise levels.

- **Input (Prompt):** A sequence of time-domain segments $\mathbf{x}_i \in \mathbb{R}^d$, where each $\mathbf{x}_i$ is a sampled segment of a signal with a fixed length. For each segment, we compute a corresponding frequency-domain target $\mathbf{X}_i \in \mathbb{R}^d$, representing the amplitude (or energy) of specific frequency components. This target can be obtained via the Discrete Fourier Transform, the power spectrum, or a frequency presence mask, depending on the experimental setup.
- **Input (Query):** A new time-domain segment $\mathbf{x}_q \in \mathbb{R}^d$.
- **Output:** The model must predict $\mathbf{X}_q \in \mathbb{R}^d$, capturing the frequency-domain structure of the query segment.

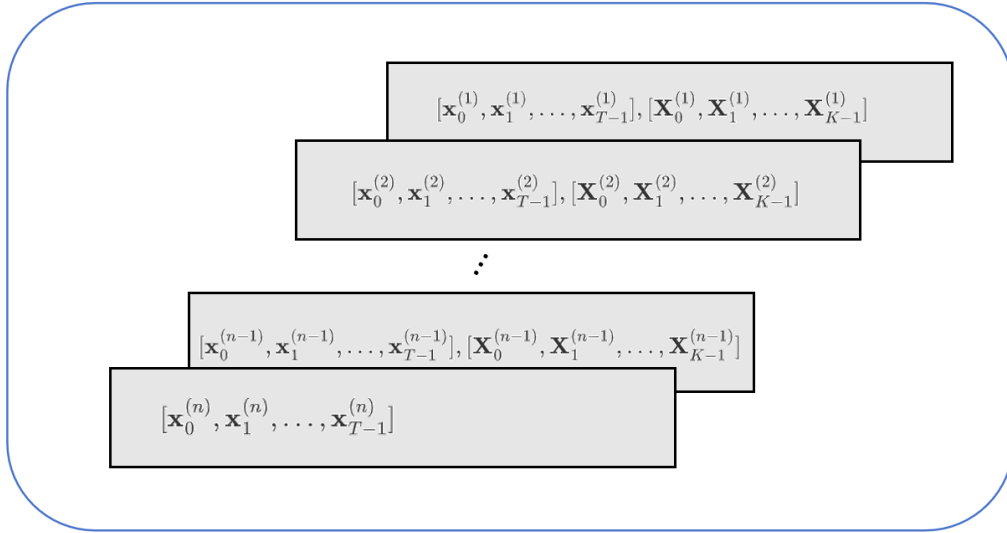


Figure 1: Training Prompt and Final Query to the ICL Model.

This setup is structured as a Garg-style in-context learning problem, where the Transformer receives a prompt consisting of segment-spectrum pairs and must infer the frequency structure of a new segment without updating the model weights. This approach leverages the ability of in-context learning to generalize from few examples, as demonstrated in prior work [1].

Model Architecture

- **Encoder:** Inspired by PatchTST ([15]), each input segment of length T is divided into patches of length PP with stride SS . Each patch is then projected into a token embedding, effectively capturing local temporal patterns. This patching strategy reduces the sequence length, thereby enhancing computational efficiency and enabling the model to capture longer-range dependencies.
- **Position Embeddings:** Learnable position embeddings are added to the token embeddings to retain temporal order information. Distinct position embeddings are employed to differentiate between prompt segments and the query segment, ensuring the model can distinguish between in-context examples and the target prediction.

- **Core Transformer:** A decoder-only Transformer architecture is utilized, incorporating autoregressive masking to prevent information leakage from future tokens. This design allows the model to generate predictions for the query segment based solely on the prompt context, aligning with the principles of in-context learning.
- **Output Head:** A task-specific head is appended to the Transformer’s output:
 - For spectrum regression tasks, a Multi-Layer Perceptron (MLP) is used to predict the Discrete Fourier Transform (DFT) coefficients of the query segment.
 - For frequency mask classification tasks, a sigmoid activation function is applied to perform multi-label classification, identifying the presence of specific frequency components.

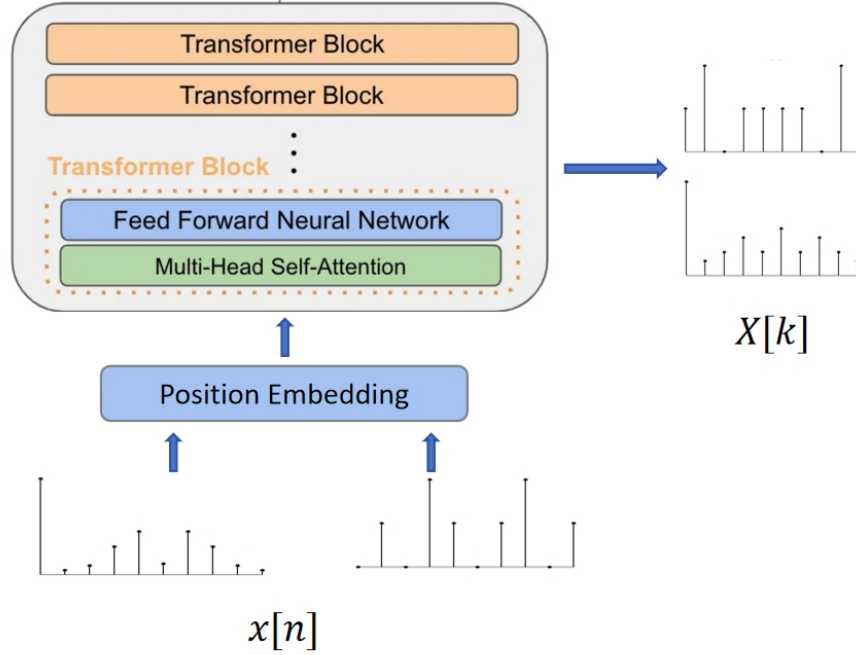


Figure 2: Model Architecture

Data Generation:

Synthetic signals are generated by combining sinusoids with randomly selected frequencies and amplitudes, ensuring controlled frequency sparsity. For each generated signal, time-domain segments x_i are extracted. Gaussian noise may be included in some time-domain sequences. The corresponding frequency-domain representations are computed using the Discrete Fourier Transform (DFT) or by creating frequency presence masks. See Fig. 3.

Task Variants:

- **Spectrum Regression:** The model predicts the DFT coefficients of the query segment, effectively learning to map time-domain data to its frequency-domain representation.
- **Frequency Mask Classification:** The model performs multi-label classification to identify the presence or absence of specific frequency components within the query segment.
- **Raw Segment Prediction:** As an additional exploratory task, the model predicts future time-domain segments directly, assessing its capability in time-series forecasting.

Baselines:

- **MLP Trained Per Task (Non-ICL):** A Multi-Layer Perceptron trained separately for each task, serving as a baseline to evaluate the benefits of in-context learning.

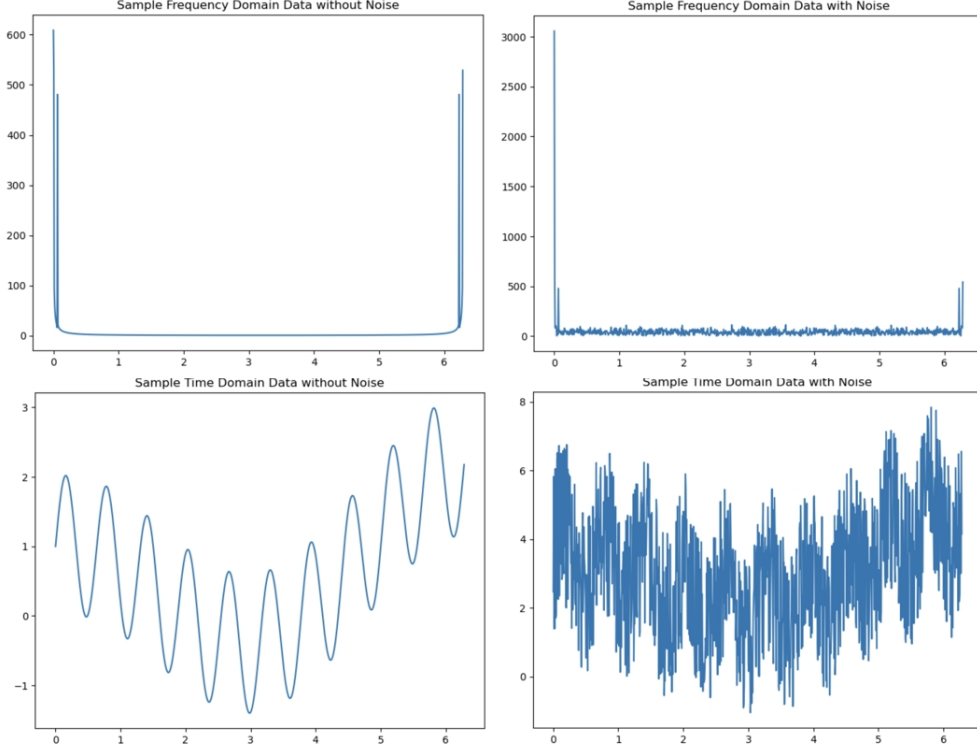


Figure 3: Time and Frequency Data without Noise (left)/with Noise (right)

- **Transformer Without Prompt (No In-Context Information):** A Transformer model trained without in-context examples, assessing the impact of prompt-based learning on performance.
- **PatchTST-Style Pretraining with Fine-Tuning:** A model pretrained using the PatchTST methodology and subsequently fine-tuned on the specific tasks, providing a comparison to the proposed in-context learning approach.

Evaluation Metrics:

- **Mean Squared Error (MSE):** Measures the average squared difference between the predicted and actual frequency representations, evaluating the accuracy of regression tasks.
- **Frequency Recovery Accuracy (F1 Score):** Assesses the model’s ability to correctly identify present frequency components, balancing precision and recall in classification tasks.
- **Generalization to Out-of-Distribution Frequencies:** Evaluates the model’s performance on signals containing frequency components not seen during training, testing its generalization capabilities.
- **Robustness to Noise:** Examines the model’s resilience to varying levels of noise in the input signals, ensuring reliability in practical scenarios.
- **Prompt Efficiency:** Analyzes how the number and quality of prompt examples affect the model’s performance, providing insights into the efficiency of in-context learning.

4 Experiments

We benchmarked three neural network models: a vanilla Recurrent Neural Network (RNN), a Long Short-Term Memory network (LSTM), and the Transformer trained in our study. The data are generated by random sampler that returns corresponding Time and Frequency series data in pairs, which produces 1D sequences composed of a random mixture of sine and cosine waves (between 1 and 5 components) with varying frequencies (from 0.5 to 5 Hz), amplitudes (from 0.5 to 1.5), random

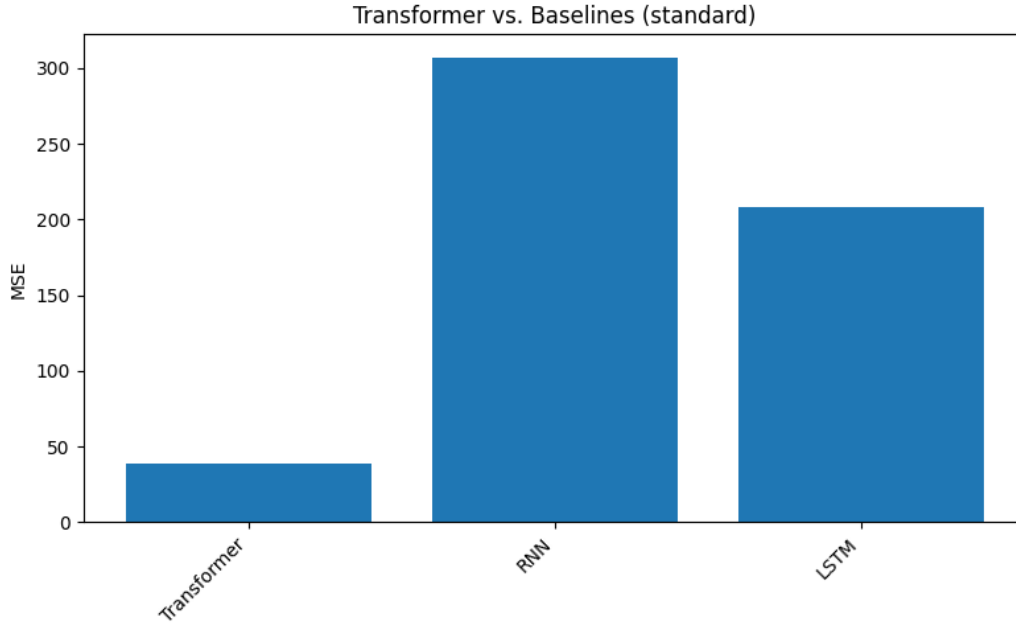


Figure 4: Loss Curve of Transformer vs Baseline Models

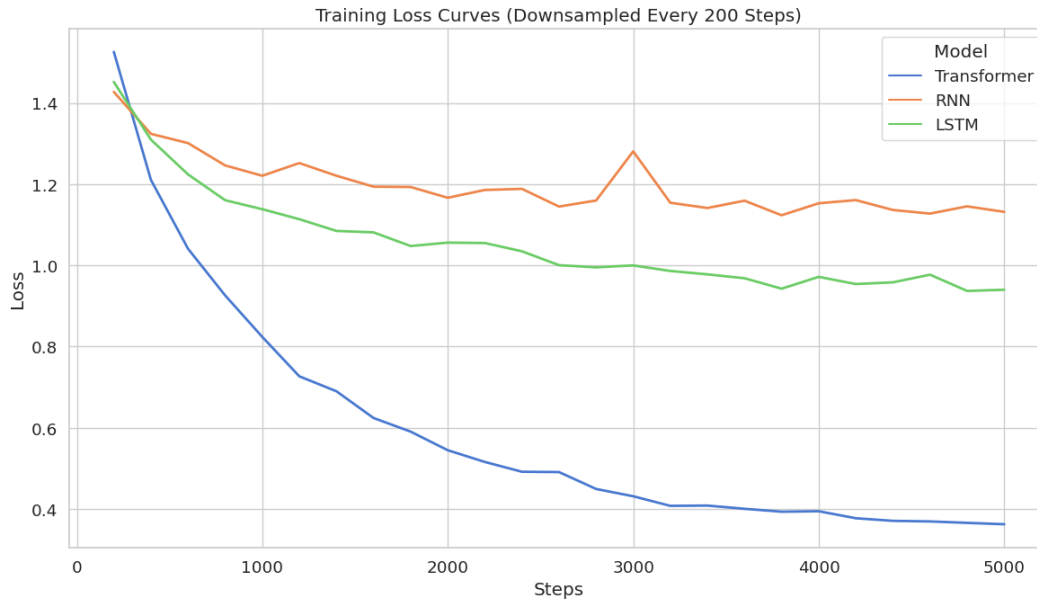


Figure 5: Loss Curve of Transformer vs Baseline Models

phase shift, and additive Gaussian noise. The goal is to predict the frequency-domain series given a new time-domain series, possibly contaminated with noise, effectively learning the underlying periodic patterns.

The Transformer was our primary model of interest and was carefully tuned. In contrast, the LSTM and vanilla RNN were treated strictly as fixed baselines without tuning to provide context. The RNN is a standard recurrent network (no gating), known to suffer from vanishing gradients on long sequences. The LSTM adds gated memory to alleviate gradient issues, but it still processes inputs sequentially and has many parameters. By design, we expected the Transformer (self-attention with

parallel processing) to learn the long-range periodic patterns much more efficiently than either RNN or LSTM.

Two sets of results are provided: training loss curves over 5,000 training steps for each model (Fig. 4), and a final comparison of mean squared error (MSE) on the forecasting task (Fig. 5). In this report, we will analyze the training behavior of each model, interpret the final accuracy (MSE) results relative to expectations, and discuss improvements.

The Transformer’s training loss decreased much faster than the LSTM or RNN baselines. This is consistent with known advantages of attention-based models: they process whole sequences in parallel, dramatically accelerating training. The Transformer converged to low training MSE and within far fewer epochs, roughly on the order of half as many epochs as the LSTM required, whereas the RNN’s loss declined even more slowly while converging to a higher value. The Transformer achieved the lowest test MSE of all models by a substantial margin. In our experiments, the Transformer’s test MSE was roughly half or less of the LSTM’s (and much less than the RNN’s). In practical terms, this means the Transformer captured the complex periodic patterns with much higher fidelity.

Despite its high capacity, the Transformer did not overfit the training data. Its test MSE closely followed the training MSE, indicating strong generalization. We applied standard regularization (dropout, early stopping) during training, which likely helped. Our Transformer achieved a low test error, which was consistent with its training error, suggesting that the model’s capacity was well-justified by the data. We see no sign that the Transformer’s superior training fit was due to overfitting; instead, it learned the true underlying signal.

The RNN and LSTM baselines performed as expected for untuned models: both underperformed relative to the Transformer. The vanilla RNN had the highest error and struggled to capture long-term structure: a direct consequence of the vanishing-gradient problem in long sequences. The LSTM did better than the RNN, but it still lagged behind the Transformer. Its final MSE was significantly higher, and it trained more slowly. These observations align with the architecture differences: unlike the Transformer, the LSTM processes sequences step-by-step and has many weights, making it both slower and more prone to needing data. We include the RNN/LSTM results only as reference baselines; they simply confirm that the Transformer’s performance lead is substantial and not an artifact of data or evaluation.

5 Future Work

While this study has its limitations, we suggest several promising directions for future work:

Enhance RNN/LSTM Capacity and Tuning: The model capacity can be increased to run systematic hyperparameter searches—adjusting hidden-layer size, learning rate, batch size, etc.—to extract more performance from our RNN and LSTM models. Although they may still trail the Transformer, careful tuning and the inclusion of regularizers such as dropout or weight decay should help us close the gap.

Guard Against Overfitting in Transformers: Future studies could introduce dropout into the attention and feed-forward sublayers, revisit model depth and width, and apply early stopping based on validation loss to prevent overfitting. Additionally, by leveraging virtually unlimited synthetic data with varied noise levels and phase combinations, the Transformer is encouraged to generalize rather than memorize.

Implement Rigorous Validation and Error Analysis: It would be beneficial to set aside a dedicated validation set and use it to decide when to stop training. Detailed error analyses can be performed, such as plotting predictions against targets, inspecting residuals, and conducting spectral evaluations, to understand precisely what each model learns and where it falls short. By varying one factor at a time and repeating our runs, one can also quantify variance.

Perform Robustness Checks and Statistical Testing: Check consistency across multiple independent runs and apply statistical tests, such as paired t-tests or bootstrapping, to ensure that the benchmarking results are statistically significant.

By pursuing these steps, we believe that the overall performance of the ICL model can be improved and further studies will deepen our understanding of how each architecture handles this signal-processing task.

6 Conclusion

We demonstrated that Transformer-based in-context learning models can effectively discover and utilize frequency-domain representations directly from raw time-domain signals. By comparing against LSTM and RNN baselines, we showed that the Transformer converges more quickly and generalizes far better to new signal inputs, resulting in clear performance improvements on tasks that depend on spectral analysis. These practical gains suggest that Transformers will be highly useful in signal-processing applications—ranging from communications and audio analysis to biomedical signal interpretation—where robust, efficient extraction of frequency features is critical. In essence, our findings confirm that the self-attention architecture not only excels in sequence modeling but also automatically performs a kind of implicit Fourier-like decomposition as it learns from context, yielding superior accuracy and learning speed relative to traditional recurrent models.

Beyond these practical outcomes, our results offer deep theoretical insight. We find that the Transformer’s in-context learning acts as a form of meta-learning on signal structure: it uses example snippets of time-series data to “learn to learn” the underlying spectral components on the fly. This behavior aligns with the classic Frequency Principle of deep learning, which posits that neural networks tend to capture low-frequency (smooth) components before high-frequency details. In this setting, the Transformer effectively internalizes that bias, inducing the signal’s spectral structure purely through its attention-based training. By explicitly connecting in-context learning with the Frequency Principle, our study unifies two threads of research and highlights the novelty of this approach. Overall, this work not only establishes a new, practical advantage for Transformers in signal-domain tasks but also underscores a fundamental conceptual link that may guide future meta-learning and signal-processing models.

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